FINANCIAL MANAGEMENT
Suggested Answers
March-April 2022

Answer to the Question # 1 (a)

Factoring services evaluation – Craft Catering

Projected annual credit sales =	Tk 8,000,000×0.8	= 6,400,000
Factoring service:		
Fees	Tk 6,400,000 × 1.5%	= 96,000
Financing	Tk 6,400,000×0.75×(30/365)×0.09	= 35,507
Annual cost		131,507
Bank overdraft*	Tk 6,400,000×0.25×(30/365)×0.08	= 10,521
		142,028
Less: Administrative savings		20,000
Net cost of factoring		122,028

Present arrangement:

Bank overdraft	= (Tk 6,400,000×70/365)×0.08	=88,192	
Bad debts	= Tk 6,400,000×0.01	=64,000	162,192
Net benefit of factoring			40,164

**With a factoring service the business will still need to finance 25 pe cent of its credit sales for 30 days.*

By entering into a factoring arrangement, the company would save Tk 40,164 per year.

With an overdraft the company would need to be able to borrow Tk 1,227,397 (Tk 6,400,000 ×70/365) at an interest rate of 8 per cent. This is a very substantial amount for a small enterprise and may well be beyond the company's borrowing capacity. The bank would inevitably require security for a loan of this amount. This may not be available, or if it is, John may wish to reserve it for other purposes.

With a factoring arrangement the company would seem to need access to borrowings of Tk 131,507 [Tk 6,400,000×0.25×(30/365)]. For a company of this size this would appear to be a more reasonable amount.

Answer to the Question # 1 (b) (i)

Evaluation of dividend policies – Energy Systems

In the scenario presented, the dividend per share (DPS) and payout ratios are calculated as follows:

Year	Forecast EPS (Tk)	DPS (Tk) Policy		Payout ratio % Policy	
		1	2	1	2
Current	9.80	3.92	3.92	40	40
1	11.90	3.92	4.70	40	40
2	10.10	4.76	5.64	40	56
3	13.30	4.04	6.77	40	51
4	14.50	5.32	8.13	40	56
5	17.40	6.96	9.75	40	56

The dividend per share for the current year is calculated as Tk 3.92 (Tk 1.02 interim plus Tk 2.09 final). With earnings per share (EPS) of Tk 9.80 expected in the current year, the payout ratio for the current year would be 40 per cent (Tk3.92/Tk 9,80).

Policy 1. If the company adopted policy number 1 and maintained a constant 40 per cent payout ratio over the five years, the dividend is forecast to fall from Tk 4.76 in year 1 to Tk 4,04 in year 2, a reduction of 15 per cent. The reduction in dividend per share follows directly from the projected fall in earnings per share and would conflict with the managing director's objective of avoiding any cut in dividends. No information is given on the reason for the forecast fall in earnings, although this should be available internally to' the board.

Policy 1 highlights one of the drawbacks of following a constant dividend payout ratio - dividend payments will fluctuate directly in line with earnings. Payouts will increase in years of high earnings and fall in years of lower earnings. A dividend payout which is likely to fluctuate in this way may not appeal to some of the company's shareholders.

Policy 2. With policy number 2, maintaining a compound growth rate in dividends of 20 per cent per year might seem ambitious and demanding in view of the company's growth aspirations.

However, based on the information provided, the policy is achievable over the planning period. In addition, should the company adopt policy number 2, no reduction in dividend payments is forecast. This would comply with the managing director’s objective.

It is noted that sustaining an annual growth rate of 20 per cent in dividends will significantly increase the payout ratio (up to 56 per cent for three of the five years) and lower retentions. The dividend cover would be reduced from 2.53 times in years 1 and 2 to 1.78 times in years 4 and 5. While a cover ratio of 1.78 might still be considered adequate, some of the directors and other investors may not be comfortable with such a reduction.

Such a policy may please some shareholders (e.g., financial institutions) and perhaps raise their expectations about future dividend payments, but there is no information provided on the structure of share ownership.

The directors should really consider if this policy is affordable in relation to the company’s investment needs. The company is reported to be growing rapidly and has ambitious growth plans. How are these to be financed? Unfortunately, this cannot be determined, as there is no information given concerning the company’s investment plants.

Based on the information provided, which is limited, it appears the directors could feasibly adopt either policy. For policy number I they could simply adjust the payout ratio in year 2 to maintain a consistent and increasing dividend per share payment. Adopting policy 1 could leave the company with higher internal financing than policy 2 and may be more appropriate in relation to its strategic growth objectives.

Without further information on shareholder clienteles and company investment plans, it is not possible to arrive at a definitive policy recommendation.

Answer to the Question # 1 (b) (ii)

The directors are implying a residual approach to dividend policy, where the amount of dividend is being determined as a residual after the company’s investment financing requirements have been met.

Maximum

	EPS(Tk)	Investment financing	Dividend (Tk)	DPS (Tk) Policy		Variance Policy	
Year				1	2	1	2
Current	9.80	5.88	3.92	3.92	3.92	0	0
1	11.80	8.33	3.87	4.76	4.70	-1.19	-1.13
2	10.10	7.07	3.03	4.04	5.64	-1.01	-2.61
3	13.30	9.31	3.99	5.32	6.77	-1.33	-2.78
4	14.50	7.25	7.25	5.80	8.13	1.45	-0.88
5	17.40	8.70	8.70	6.96	9.75	1.74	-1.05

For the current year, the maximum dividend is the dividend actually proposed, and investment financing is the EPS of Tk 9.80 less the proposed dividend of Tk 3.92.

With the exception of the current year, the company's investment needs have been calculated as a percentage of the forecast earnings per share. For example, in year 1 the company's investment needs are equivalent to Tk 8.33 per share (Tk 11.90). This would leave a maximum residual dividend of Tk 3.57 per share available in year 1. Whether the company would wish to use all of this for dividends and leave no reserves is another matter. This would have implications for the company's capital structure.

Adopting a purely residual approach will produce an initial decline in the pattern of dividend payments, from the current year's proposed dividend of Tk 3.92 through to the dividend in year 2 of Tk 3.03. The level of dividend per share is not expected to return to current levels until year 4. After which time, there would seem to be scope for a substantial increase in dividends.

Shareholders may not react favorably to a decline in their dividends. Although, if they are informed of the company's plans, they may be prepared to accept an initial dividend reduction if they are persuaded this policy will produce higher future dividends through the profitable investment of the retained earnings.

In relation to the two dividend policies originally proposed, it is evident that the directors would not be able to adopt either policy, given the constraint on additional financing. With policy 1, financing would be needed during years 1 to 3, and policy 2 could not be achieved without significant additional financing over the five years.

Answer to the Question # 2

Costs of Capital

Cost of Equity (using CAPM) = Rf + [Ba x (Rm - Rf)]

Rf = 2%
Rm = 10%
Ba = 1.2

Cost of Equity (using CAPM) = 2% +[1.2 x (10%- 2%)]
= 2% +[1.2 x (8%)]
= 2% +[0.096]
= 11.60%

The 6% irredeemable debentures:

The yield on this can be estimated solving for Kd in the following perpetuity formula: Po = I / Kd
Kd = the <u>after tax</u> cost of debt
Note: tax of 12.5% must be deducted from the interest payments.

i.e. interest is BDT 3.00 per nominal BDT 100, every six months

Hence after tax payment = BDT 3.00 (1-0.125) = 2.63 per nominal BDT 100,

Po = I / Kd where: Po = BDT 110.0and I = BDT 2.63
=> Kd = I / Po = 2.625 / 110 = 2.39% semiannual = **4.77 % annually**

Cost of preference shares

Its preference shares has a BDT 10.00 nominal value
Dividend on the preference shares is 7%
Current market price of the preference shares is BDT 12.50
Cost of preference shares is = Actual Dividend / current market price
= (7% x BDT 10.00) / BDT 12.50
= (0.70) / 12.50
= 0.056 = **5.60%**

Market Values of the Capital Structure

The market value of Equity

Current cum div share price	BDT 48.00
Current numbers of shares	500,000
Expected dividend	BDT 3,000,000
Expected dividend per share	BDT 6.00
Current Ex div share price	BDT 42.00
Current Equity Market value	**BDT 21,000,000**

The market value of the irredeemable Debt

= the current market price, (ex interest) per bond x # of bonds issued

The 6% irredeemable debentures

= BDT 110.00 x (BDT 12,000,000 / 100) =

BDT 110.00 x (120,000) = **BDT 13,200,000**

The market value of the Preference Shares

= the current market price, (ex div) per share x # of shares issued

= 12.50 x (10,000,000 / 10.00)
= 12.50 x (1,000,000)

BDT 12,500,000

In Summary	Cost	Market Value
Ordinary Shares	11.60%	BDT 21,000,000
Irredeemable Debt	4.77%	BDT 13,200,000
Preference Shares	5.60%	BDT 12,500,000
		BDT 46,700,000

Hence the WACC = Ke_g x {E / (E + D+PS)} + Kd x {D / (E + D+PS)}+ Kps x {PS / (E + D+PS)}

= 11.60% x (BDT 21,000,000 / BDT 46,700,000)
+ 4.77% x (BDT 13,200,000 / BDT 46,700,000)
+ 5.60% x (BDT 12,500,000 / BDT 46,700,000)
= 0.05216 + 0.0135 + 0.015
= 0.08063 = **8.06%**

Alternatively	**After Tax Cost**	**Market Value (BDT)**	**Number Issued**	**Total Value (BDT)**	**Proportion**	**% Return**
Ordinary Shares	11.60%	42.00	500,000	21,000,000	45%	5.22%
Irredeemable Debt	4.77%	110.00	120,000	13,200,000	28%	1.35%
Preference Shares	5.60%	12.50	1,000,000	12,500,000	27%	1.50%
				46,700,000	**100%**	**8.06%**

Answer to the Question # 3 (a)

(a) **MEMORANDUM**

To: Board of Directors
From: Finance Team Member
Date: 23 March 2022

Forward exchange contract
20 June 2022 receipt
€632,200 / (€ 1.445 - € 0.0081) = £ 439,975

Money market hedge
20 June 20X7 receipt
€632,000 / 1+ (5.2% / 4) = € 623,889 borrowed

€ 623,889 / 1.445 = £ 431,757 received now

Note: £ 431,757 (1+(4.8%/4))

= £ 436,938 in 6 months

22nd/23rd September net payment
Netting off
Payment due = (€1,347,500)
Receipt due = €560,000
Net payment due = (€787,500)
€787,500 / (1.412 - € 0.0143)
€787,500 / €1.3977
= £ 563,425

Money market hedge
22nd/23rd September net payment
€787,500 / (1 + (3.9%/2)
= €787,500/ 1.0195
= €772,437
€772,437 / 1.412
= £ 547,052 paid now

Note: £ 547,052 (1+(6.1%/2))
= £ 563,737 in 6 months

For the June receipt Attire Ltd. would be better off with a forward exchange contract, as it would receive more Taka than with a money market hedge.

For the September net payment, a forward exchange contract produces a lower net payment in Taka.

	£
In summary June receipt (forward exchange contract)	439,835
September net payment (forward exchange contract)	(563,425)
Net payment	(123,590)

Were Attire Ltd. to ignore hedging then the situation, using the spot rates, would be:

June receipt (€632,000/1.445)	437,370
September net payment (€787,500/1.412)	(557,720)
Net payment	(120,350)

Based on these figures, the hedging would cost the company £ 3,240 (£ 120,350 - £ 123,590).

Answer to the Question # 3 (b)

(b) **A futures contract** is similar to a forward exchange contract (FEC) as the company would be in a no win/no loss position. However, the future is for a standardised amount, unlike an FEC, which can be for any amount. Also the company would not be able to buy a future at the bank (as it could with an FEC), as futures are traded on currency exchanges.

Currency options are similar to FECs, but they would give Attire Ltd. the *right* to buy/sell currency in the future, whereas FECs are a contractual *obligation*.

There are two types of option:

- Call option to buy currency or
- Put option – to sell currency

Because the holder of the option has the *right* to buy/sell, the option is more flexible. As a result of this options are more expensive and transaction charges are high.

Answer to the Question # 4 (a)

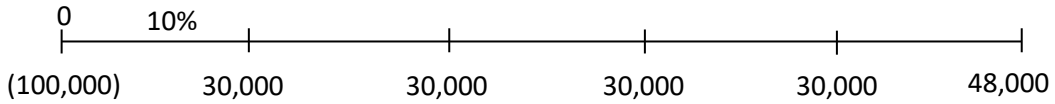
Investment decisions will be influenced by the investor's risk propensity or the investor's attitude to risk. Investors who have a low-risk propensity, in other words they have a preference for less risk, are said to be risk averse. Whereas investors who have a high-risk propensity or a positive desire for risk, are referred to as risk-taking or risk-seeking. Other individuals may be risk-indifferent, or risk-neutral, that is, for an increase in risk, they do not necessarily require an increase in return.

Before making investment decisions the financial manager will need to evaluate the risk-return characteristics of each potential investment and how they are likely to affect the risk-return characteristics of the firm overall. If investments are considered to increase the firm's risk then shareholders and other investors, who are generally considered to be risk-averse, will require a commensurate increase in return.

Answer to the Question # 4 (b) (i)

First, find the expected cash flows:

YEAR	EXPECTED CASH FLOWS			
0	0.2(-Tk100,000) +	0.6(-Tk100,000) +	0.2(-Tk100,000) =	(Tk100,000)
1	0.2(Tk20,000) +	0.6(Tk30,000) +	0.2(Tk40,000) =	Tk30,000
2				Tk30,000
3				Tk30,000
4				Tk30,000
5				Tk30,000
5*	0.2(Tk0) +	0.6(Tk20,000) +	0.2(Tk30,000) =	Tk18,000



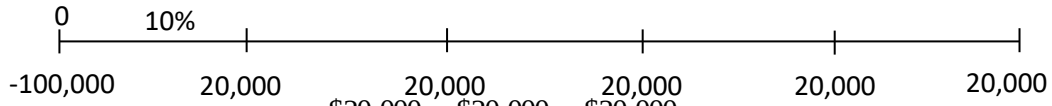
Next, determine the NPV based on the expected cash flows:

$$NPV = Tk100{,}000 + \frac{\$30{,}000}{(1.10)^1} + \frac{\$30{,}000}{(1.10)^2} + \frac{\$30{,}000}{(1.10)^3} + \frac{\$30{,}000}{(1.10)^4} + \frac{\$48{,}000}{(1.10)^5} = \$24{,}900\,.$$

Using a financial calculator, input the cash flows in the cash flow register, enter I = 10, and then press the NPV key to obtain NPV= Tk24,900.

Answer to the Question # 4 (b) (ii)

For the worst case, the cash flow values from the cash flow column farthest on the left are used to calculate NPV.



$$NPV = Tk100{,}000 + \frac{\$20{,}000}{(1.10)^1} + \frac{\$20{,}000}{(1.10)^2} + \frac{\$20{,}000}{(1.10)^3} + \frac{\$20{,}000}{(1.10)^4} + \frac{\$20}{(1.10)^5} = \$24{,}184$$

$$NPV = Tk100{,}000 + \frac{Tk\ 30{,}000}{(1.10)^1} + \frac{Tk\ 30{,}000}{(1.10)^2} + \frac{Tk\ 30{,}000}{(1.10)^3} + \frac{Tk\ 30{,}000}{(1.10)^4} + \frac{Tk\ 48{,}000}{(1.10)^5} = Tk\ 24{,}900$$

Similarly, for the best case, use the values from the column farthest on the right. Here the NPV is Tk70,259. If the cash flows are perfectly dependent, then the low cash flow in the first year will mean a low cash flow in every year. Thus, the probability of the worst case occurring is the probability of getting the Tk20,000 net cash flow in each year can be low, high, or average, and the probability of getting all low cash flows will be 0.2(0.2) (0.2)(0.2)(0.2) = 0.25 = 0.00032 = 0.032%.

Answer to the Question # 4 (b) (iii)

The base case NPV is found using the most likely cash flows and is equal to Tk26,142. This value differs from the expected NPV of Tk24,900 because the Year 5 cash flows are not symmetric. Under these conditions, the NPV distribution is as follows:

P_R	NPV
0.2	(Tk24,184)
0.6	26,142
0.2	70,250

Thus, the expected NPV is 0.2(-Tk24,184) + 0.6(Tk26,142) + 0.2(Tk70,259) = Tk24,900. As is generally the case, the expected NPV is the same as the NPV of the expected cash flows found in part a. The standard deviation is Tk29,904.

$\sigma 2$ NPV =0.2(-Tk24,184 – Tk24,900)2 + 0.6(Tk26,142 – Tk24,900)2
+ 0.2 (Tk70,259 – Tk24,900)2
= Tk 894,261,126.
$\sigma 2$ NPV = $\sqrt{Tk\ 894{,}261{,}126} = Tk\ 29{,}904$.
The coefficient of variation, CV, is Tk29,904/Tk24,900 = 1.20.

Answer to the Question # 4 (b) (iv)

Because the project's coefficient of variation is 1.20, the project is riskier than average, and hence the project's risk-adjusted required rate of return is 10% + 2% = 12%. The project now should be evaluated by finding the NPV of the expected cash flows, as in part a, but using a 12 percent discount rate. The risk adjusted NPV is Tk18,357, and therefore the project should be accepted.

Answer to the Question # 5 (a)

Price or Range of Prices to Purchase Nice Home Ltd., based on P/E Ratios of Competitors:

	Tk. '000'
Earnings After Tax of Nice Home Ltd.	= 56,250
Value of Nice Home Ltd. at P/E Ratio of 14:1	= EAT x P/E Ratio
	= 56,250 x 6 ÷ 1
	= 337,500

Price per share	= 337,500 ÷ 4,500 **= Tk. 75 per share**
Value of Nice Home Ltd. at P/E Ratio of 15:1	= EAT x P/E Ratio = 56,250 x 7 ÷ 1 = 393,750
Price per share	= 393,750 / 4,500 **= Tk. 87.50 per share**

The Fine Home Ltd., can offer the current market price of Tk. 65 per share to purchase Nice Home Ltd., but this price may not be acceptable to the shareholders of Nice Home Ltd. However, based on P/E ratio of competitors, having same business risk, the Fine Home Ltd., can offer any price between Tk. 75 per share to Tk. 87.5 per share.

Answer to the Question # 5 (b)

Cost of Equity:

Cost of equity (Ke) $= R_f + (R_m - R_f)$

$=$ 5+ (11-5) 1.25

$= 12.5\%$

Cost of Debt:
Assuming Fine Home Ltd's., cost of debt will remain unchanged on acquisition of Nice Home Ltd.:

Years	Cash Flow	PVF @ 5%	PV	PVF @ 6%	PV
0	(105)		(105)	1.000	(110)
1 to 3	6.75*	2.723	18.38	2.673	18.04
4	106.75	0.823	87.86	0.792	84.54
			1.24		(2.41)

IRR = 5 + ((1.24/ (1 .24 + 2.41)) X (6 -5) = 5.34%

*After tax annual interest expense = (100 X 9%) X (1- 25% tax) = 9 X 75% = 6.75

Weighted Average Cost of Capital:
WACC at 30% gearing level $= (W_d \times K_d) + (W_e \times K_e)$
$= (0.3 \times 5.34) + (0.70 \times 12.5)$
$= 1.60 + 8.75$
= 10.35%

WACC at 40% gearing level = (0.40 x 5.3) + (0.60 x 16.68)
= 2.14 + 7.5
= 9.64 %

Answer to the Question # 5 (c)

Future Cash Flows: Tk. '000'

	2020-21	**2021-22**	**2022-23**	**2023-24**	**2024-25**
Sales	550,000	605,000	653,400	686,000	713,450
Cost of goods sold	-247,500	-272,250	-294,030	-308,700	-321,053
Gross Profit	302,500	332,750	359,370	377,300	392,398
Operating expenses	-77,000	-84,700	-85,000	-89,180	-90,000
Operating profit	225,500	248,050	274,370	288,120	302,398

Capital allowances	-81,250	-85,000	-87,500	-87,500	-87,500
Profit before tax	144,250	163,050	186,870	200,620	214,898
Tax @ 32%	-46,160	-52,176	-59,798	-64,198	-68,767
Profit after tax	98,090	110,874	127,072	136,422	146,130
Capital allowances	81,250	85,000	87,500	87,500	87,500
Working capital	-85,000	-82,000	-85,000	-90,000	-95,000
Net cash flows	94,340	113,874	129,572	133,922	138,630

Answer to the Question # 5 (d)

	2020-21	2021-22	2022-23	2023-24	2024-25
Present value at 5% growth:					
Net cash flows	94,340	113,874	129,572	133,922	138,630
Terminal cash flow (5% growth)W-1					2,426,025
PVF@ 11%	0.901	0.812	0.731	0.659	0.593
Present Values	85,000	92,466	94,717	88,254	1,520,841
PV	**1,881,278**				
Present value at 7% growth:					
Net cash flows	94,340	113,874	129,572	133,922	138,630
Terminal cash flow (7% growth) W-2					3,708,353
PVF@ 11%	0.901	0.812	0.731	0.659	0.593
Present Values	85,000	92,466	94,717	88,254	2,281,261
PV	**2,641,698**				

W-1: Terminal cash flow (5% growth): (Amount in Tk. '000)

$$= \frac{\text{2024-25 earnings } (1+g)}{(r-g)}$$

$$= \frac{138{,}630(1+0.05)}{(0.11-0.05)}$$

$$= 2{,}426{,}025$$

W-2: Terminal cash flow (7% growth):

$$= \frac{\text{2024-25 earnings } (1+g)}{(r-g)}$$

$$= \frac{138{,}630 \quad (1+ \quad 0.07)}{(0.11-0.07)}$$

$$= 3{,}708{,}353$$

Answer to the Question # 5 (e)

Value of Shares:
At 5% growth:
Value of shares at 30% gearing = Tk. 1,881,278 x 70% = Tk. 1,316,895 (292.64 Tk. Per share)
Value of shares at 40% gearing = Tk. 1,881,278 x 60% = Tk. 1,128,767 (250.84 Tk. Per share)
Value of shares will be between Tk. 1,128,767 and Tk. 1,316,895

At 7% growth:
Value of shares at 30% gearing = Tk. 2,641,698 x 70% = Tk. 1,849,189 (410.93 Tk. Per share)
Value of shares at 40% gearing = Tk. 2,641,698 x 60% = Tk. 1,585,019 (352.23 Tk. Per share)
Value of shares will be between Tk. 1,585,019 and Tk. 1,849,189

Answer to the Question # 6 (a)

Tk100,000,000/10 = Tk10,000,000 per year, or Tk5 million each 6 months. Because the Tk5 million will be used to retire bonds immediately, no interest will be earned on it.

Answer to the Question # 6 (b)

The debt service requirements will decline. As the amount of bonds outstanding declines, so will the interest requirements (amounts given in millions of Taka):

Semiannual Payment Period (1)	Sinking Fund Payment (2)	Outstanding Bonds on which Interest is Paid (3)	Interest Payment (4)	Total bond service (2)+(4)=(5)
1	Tk5	Tk100	Tk6.0	Tk11.0
2	5	95	5.7	10.7
3	5	90	5.4	10.4
:	:	:	:	:
20	5	5	0.3	5.3

Interest s calculated as [(0.12).2] (Column 3); for example: interest in Period 2 = (0.06) Tk95) = Tk5.7.

The company's total cash bond service requirement will be Tk21.7 million for the first year. The requirement will decline by 0.12 (Tk10,000,000) = Tk1,200,000 per year for the remaining years.

Answer to the Question # 6 (c)

If interest rates rose, causing the bond's price to fall, the company would use open market purchases. This would reduce its debt service requirements.

---The End---